

Yonghyun Kim

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Current Research Topics

Currently, I am interested in researching (1) verified compilers, (2) software formal verification, and (3) property-based testing of security properties.

Research Interests

Compiler Verification; Compiler Optimization; Software Verification; Speculative Execution Defenses; Information Flow Control

Skills

Compiler; Formal Method (Formal Verification); Coq

Positions

2025.3-2025.4 Research Assistant, Seoul National University Institute of Computer Technology
2025.5- Postdoctoral Researcher, Max Planck Institute for Security and Privacy
Supervisor: [Cătălin Hrițcu](#)

Education

2017.3-2025.2 Ph.D student, Computer Science and Engineering, Seoul National University
Supervisor: [Chung-Kil Hur](#)
2009.2-2016.8 B.S. Student, Computer Science & Mathematical Sciences (double major), KAIST
2007.3-2009.2 Daejeon Science High School

Grants, honors & awards

2009-2013 Republic of Korea Presidential Science Scholarship, Korea Student Aid Foundation.
2026.5-2027.4 Global Advanced Cybersecurity Human Resources Development, Institute of Information & Communications Technology Planning & Evaluation(IITP), Republic of Korea

Publications

CONFERENCES

AI Engram: In Search of Memory Traces in Artificial Intelligence

Jea Kwon, Dong-Kyum Kim, Jiwon Kim, Yonghyun Kim, Woong Kook, and Meeyoung Cha
In Proceedings of the 43rd International Conference on Machine Learning 2026 (ICML 2026)

Archmage and CompCertCast: End-to-End Verification Supporting Integer-Pointer Casting
Yonghyun Kim, Minki Cho, Jaehyung Lee, Jinwoo Kim, Taeyoung Yoon, Youngju Song, Chung-Kil Hur

In Proceedings of the 52nd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2025)

VeriRT: An End-To-End Verification Framework for Real-Time Distributed Systems

Yoonseung Kim, Sung-Hwan Lee, Yonghyun Kim, Chung-Kil Hur

In Proceedings of the 52nd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2025)

CompCertM: CompCert with C-Assembly Linking and Lightweight Modular Verification

Youngju Song, Minki Cho, Dongjoo Kim, Yonghyun Kim, Jeehoon Kang, Chung-Kil Hur

In Proceedings of the 47th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2020).

Crellvm: Verified Credible Compilation for LLVM

Jeehoon Kang, Yoonseung Kim, Youngju Song, Juneyoung Lee, Sanghoon Park, Mark Dongyeon Shin, Yonghyun Kim, Sungkeun Cho, Joonwon Choi, Chung-Kil Hur, Kwangkeun Yi.

(The first three authors contributed equally and are listed alphabetically)

In Proceedings of the 39th annual ACM SIGPLAN conference on Programming Languages Design and Implementation (PLDI 2018).

PREPRINTS

Triosecuris: Formally Verified Protection Against Speculative Control-Flow Hijacking

Jonathan Baumann*, Yonghyun Kim*, Yan Farba, Catalin Hritcu, Julay Leatherman-Brooks
(* Equal contribution)

January 2026.

BOOKS

Software Foundations: Security Foundations

Cătălin Hrițcu and Yonghyun Kim.

[Electronic textbook volume](#), January 2026.

Teaching

TA, Proofs are Programs, 2026 Summer (Prof. Cătălin Hrițcu and Prof. Clara Schneidewind)

TA, Foundations of Programming Languages, Verification, and Security, 2025 Summer (Prof. Cătălin Hrițcu and Prof. Jana Hofmann)

TA, Principles and Practices of Software Development, 2017 Spring, 2018 Spring, 2019 Spring (Prof. Chung-Kil Hur)

TA, Principles of Programming, 2018 Fall, 2019 Fall, 2020 Fall (Prof. Chung-Kil Hur)

Experience

2021.9-2022.8 Student Researcher(Part Time), Google Korea, Host: [Jieung Kim](#)

Community Service

AEC, 53rd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2026)

PC, 17th Conference on Interactive Theorem Proving (ITP 2026)

PC, 2026 Workshop on Foundations of Computer Security (FCS 2026)